FLEET TEAM DIGEST

737MAX-FTD-21-21005

Issue Title : Potential Reduced Cargo Fire Suppression Capability During Dispatch of eFlow Aircraft in Pneumatic Mode

Airplane Model	ATA	Minor Model(s)
737MAX	2150-00	-7, -8, -8200, -9
Other Model(s)	**ECCN**	
737NG	9E991	

Originated Date	Last Revised Date	Created On
08/18/2021	01/05/2023	11/14/2022
Estimated Completion Date	**Next Update Date**	

Revision Description

Moved information from the Status section to the Background section.
Updated the Status and Milestones sections to reflect latest actions taken to close this project.

This FTD article is closed.

Changes are highlighted in blue text.

Applicability

All 737-7, -8, -8200, -9, -10 airplanes

Description

The FAA issued Airworthiness Directive (AD) 2021-16-15 on August 9, 2021 with a compliance date of August 19, 2021, requiring a change to the dispatch configuration guidance in the 737 MAX Master Minimum Equipment Lists/Dispatch Deviation Guide (MMEL/DDG) 21-51-02-01 when dispatching with failed, disabled, or inoperative electronic flow control of one or both air conditioning packs.

The AD prohibits the carriage of cargo in the aft cargo compartment when the airplane is dispatched or released with failed, disabled, or inoperative electronic flow control of one or both air conditioning packs. Unlike the forward cargo compartment where Boeing established the relationship between cabin air flow rate and cargo compartment leakage, additional data is needed to ensure fire suppression concentrations are maintained in the aft cargo compartment when operating with one or both packs in pneumatic mode in case of a cargo fire event. This is due to the increased cabin airflow rate on leakage being greater in the aft cargo compartment due to the proximity to the outflow valve. In contrast, the forward cargo compartment has a lower rate of leakage through the cargo liners and cargo door due its location with respect to the outflow valve, and is not impacted by the AD.

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The AD also requires the same aft cargo hold restrictions for a 1 Pack Inoperative dispatch in accordance with 737 MAX MMEL/DDG 21-51-01-01A. While FAA AD 2021-16-15 does not explicitly reference the Pack Inoperative MMEL/DDG, the FAA mandated the need for an AMOC before relieving the aft cargo carriage restriction if operators were to use the 1 Pack Inoperative MMEL/DDG for dispatch with failed electronic flow control.

NOTE: When an airplane is dispatched with one (1) inoperative air conditioning pack, there is no flow to the respective pack and there are no concerns with aft cargo compartment Halon concentration. The concern only exists when a pack is dispatched with the primary (electronic) controls inoperative and the pneumatic backup regulations (built into the flow control valve) is providing control for the pack. Test data has been collected showing that hazardous quantities of smoke or extinguishing agent will not migrate to occupied areas when operating with a single pack in primary (electronic) control.

Background

The 21-51-02-01 MMEL/DDG allows for dispatch with one or both air conditioning electronic flow control valves (eFCV) operating in pneumatic regulation for a 10-day period if electronic flow control has failed. When operating in pneumatic flow control mode, the air conditioning packs provide significantly more air than when electronically controlled in normal or smoke modes.

The cargo fire suppression agent concentration may be reduced by higher than normal air conditioning pack flow rates in pneumatic flow control mode due to increased cargo compartment leakage. The pneumatic flow control mode was not tested for fire suppression, and Boeing does not have data to show that the required 3% average Halon concentration level can be maintained in the aft cargo compartment if an aft cargo fire occurs on an aircraft when dispatched with one or both packs in pneumatic flow control mode. As such, the aft cargo compartment fire suppression capability may be reduced if the airplane is dispatched or released with failed, disabled, or inoperative electronic flow control of one or both air conditioning packs, which is currently allowed under MMEL 21-51-02-01.

For all 737 MAX airplanes, in accordance with MMEL 21-51-02-01, failure of the electronic flow control (through failure of the eFCV torque motor or primary PFTC) of one or both air conditioning packs will result in backup pneumatic flow control. Status message PACK FLOW PRIMARY L or PACK FLOW PRIMARY R will be present to confirm this condition exists.

Boeing reviewed this dispatch configuration as part of Continued Operational Safety Program (COSP) 2021-0915. As part of the COSP, the FAA completed a review of this dispatch configuration and determined that a safety issue exists and must be addressed. As a result, FAA AD 2021-16-15 was issued as described in the Description section.

No pilot action is required in-flight as AD 2021-16-15 only affects aircraft on ground prior to dispatch. The MMEL/DDG considers that the airplane is being dispatched in a degraded condition and the operational restrictions in the MMEL/DDG must address the subsequent failure of the next critical system or system functionality (next critical failure). If the airplane is dispatched with the electronic flow control inoperative, a failure of the pneumatic back-up flow control mode was the next critical failure considered during the original development of the MMEL. AD 2021-16-15 addresses a concern about adequate Halon concentrations, should a fire in the aft cargo bay occur following MMEL dispatch with the electronic flow control inoperative. To reiterate, the safety concern surrounding the AD is only applicable to the MMEL dispatch scenario.

If failure of the electronic flow control of one or both air conditioning packs occurs during flight, a Maintenance light will illuminate after landing, at which time the proper maintenance actions should be performed based on the detected faults and

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maintenance messages in accordance with standard Flight Crew Operation Manual (FCOM) protocols.

Boeing released Multi Operator Message MOM-MOM-21-0358 and hosted two (2) Out Of Sequence (OoS) Fleet Calls (FTCs) to advise operators of the AD. The OoS FTC presentation is attached for reference. A Frequently Asked Questions (FAQ) sheet based on inquiries raised during the OoS FTC has been included as an attachment under Fleet IdeaXchange (FIX) Number ISS-21-21-36790.

To provide clarity to the AD scope, Boeing submitted a Global Alternative Means of Compliance (GAMOC) request to the FAA for approval to allow unrestricted aft cargo carriage if operators apply 737 MAX MMEL/DDG 21-51-01-01A (Pack Inoperative Condition) in the event of an electronic flow system failure on one or both packs. The FAA approved the GAMOC request, which was issued as Item No. 785-21-11462 on August 13, 2021. This information has been shared in MOM-MOM-21-0358-01B(R1), issued on August 17, 2021.

Status

Updates to the 737 MAX MMEL and DDG (Item No. 21-51-02-01) were incorporated in December 2021 and January 2022, respectively. Procedures should be established to ensure the aft lower cargo compartment remains empty or is verified to contain only empty cargo handling equipment, ballast (ballast may be loaded in ULDs), and/or Fly Away kits.

Boeing initiated Service Related Problem (SRP) 737MAX-SRP-21-0282 to investigate this issue. Boeing gathered data in September 2022 to better understand the impact of the MMEL condition on cargo fire suppression performance. Based on the results of the test data, there are no plans to restore the original MMEL (no aft cargo carriage restriction) with regard to the 737 MAX fleet. With the above actions, Boeing considers the aforementioned SRP investigation closed.

Interim Action

None

Final Action

In coordination with the FAA, Boeing has updated MMEL/DDG 21-51-02-01 with revised dispatch criteria to ensure it aligns with the AD requirements. Operators should limit the allowable aft cargo loading in accordance with the aforementioned MMEL/DDGs.

Milestones

MOM-MOM-21-0358-01B released: 8-Aug-2021 (complete)

FAA AD 2021-16-15 released: 9-Aug-2021 (complete)

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GAMOC request submitted to the FAA to allow aft cargo hold carriage during 1 Pack Inoperative dispatch: 9-Aug-2021 (complete)

Out-of-Sequence (OoS) Fleet Team Calls: 10/11-Aug-2021 (complete)

FIX Item No. ISS-21-21-36790 Posted: 12-Aug-2021 (complete)

FAA GAMOC (Item No. 785-21-11462) Approval: 13-Aug-2021 (complete)

MOM-MOM-21-0358-01B(R1) released: 17-Aug-2021 (complete)

Effective Compliance Start Date for FAA AD 2021-16-15: 19-Aug-2021

Release updated MMEL/DDG 21-51-02-01 dispatch criteria: December 2021/January 2022 (complete)

Gather data on the impact of the MMEL condition on cargo fire suppression performance: End of September 2022 (complete)

Operator Action

With the electronic flow control valve in backup pneumatic flow control, a maintenance light will illuminate upon landing. The maintenance light will direct operators to use the MAX Display System (MDS) on the P9 panel. On the ONBD MAINT tab, there will be an option to choose Line Maintenance OR Extended Maintenance. The Line Maintenance tab will show the inbound flight deck effects (FDE) and existing FDE. The Extended Maintenance tab will show the present leg faults, existing faults and fault history. Status message PACK FLOW PRIMARY L or PACK FLOW PRIMARY R will be present to confirm this condition exists. If any of the above-noted Status messages are observed, the aircraft must comply with the AD ruling as noted earlier in this FTD article.

If operators choose to dispatch the aircraft under MMEL/DDG 21-51-01-01A (1 Pack Inoperative Condition), they are allowed to deviate from the AD's aft cargo carriage restriction in accordance with FAA-approved GAMOC Item No. 785-21-11462. In addition, if an operator were to apply the Pack Inoperative MMEL/DDG for a different failure condition (i.e. ACM failure for example), then compliance to the AD would not be required.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Related Problem (SRP)	737MAX-SRP-21-0282		
Airworthiness Directive (AD)	2021-16-15		
Minimum Equipment List	21-51-02-01		

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(MEL)			
Minimum Equipment List (MEL)	21-51-01-01A		
Other	GAMOC 785-21-11462		08/13/2021
Multi-Operator Message (MOM)	MOM-MOM-21-0358		
Fleet Team Safety (FTS)	ISS-21-21-36790		08/12/2021
Other	COSP 2021-0915		
Fleet Team Digest Article (FTD)	737MAX-FTD-21-21002		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Electronic Flow Control Valve	63396754-2		
Integrated Air Systems Controller	51090322-001		
Integrated Air Systems Controller	51090322-002		

Related Categories

- Regulatory : Issues mandated by airworthiness directive or other regulatory guidance material

Attachments:

a) FAQ Sheet_Revised

b) OOS Deck - Dispatch in Pneumatic Mode

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